Two-Axis Solar Tracker

Arduino control, servo movement and 3D design

Arduino Nano • LDR sensors • SG90 servos • 3D-printed structure

Group 3 – ENSA Agadir – 2025/2026

Project overview

This project focuses on the design of a two-axis solar tracker prototype. The system is based on an Arduino Nano, four light-dependent resistors (LDRs), and two SG90 servo motors. The Arduino reads the sensor values and uses them to control the horizontal and vertical movements of the structure.

Our work focused on three main areas: programming the Arduino, testing the movement of the servo motors, and designing and 3D-printing the mechanical structure.

Scope of this report. The solar panel was not used to measure electrical output during this project. No energy, efficiency, or comparison measurements were calculated, so this report does not present any energy-performance results.

Project objectives

The objectives were to:

- control two servo motors with an Arduino Nano;
- test horizontal and vertical movement;
- prepare a control method based on four LDR sensors;
- design a suitable two-axis mechanical structure in 3D;
- manufacture the structure using 3D printing.

Components

Component	Quantity	Purpose
Arduino Nano	1	Runs the program and controls the servos
SG90 servo motor	2	Moves the horizontal and vertical axes
Light-dependent resistor (LDR)	4	Measures light from four directions
10 kΩ resistor	4	Forms a voltage divider with each LDR
Breadboard and jumper wires	–	Used for circuit assembly and testing
3D-printed structure	1	Holds the motors, sensors and future panel

Arduino code

The program is written in C++ for the Arduino environment and uses the `Servo.h` library. The code begins by defining the pins, the control parameters and the initial servo positions.

Pin assignment

Element	Pin	Function
Top-left LDR	A0	Light reading v_1
Top-right LDR	A1	Light reading v_2
Bottom-left LDR	A2	Light reading v_3
Bottom-right LDR	A3	Light reading v_4
Horizontal servo	D9	Azimuth movement
Vertical servo	D10	Elevation movement

Control parameters

The current code uses the following values:

- `TOLERANCE = 50`: creates a dead zone and reduces oscillation;
- `STEP_DEG = 1`: moves a servo by one degree per correction;
- `NIGHT_THR = 100`: activates night mode when all LDR readings are low;

- `DELAY_MS = 500`: waits half a second between normal corrections.

Program sequence

In `setup()`, the two servos are attached to pins D9 and D10, moved to their initial position of 90°, and connected to the Serial Monitor at 9600 baud.

In `loop()`, the Arduino reads the four LDR values using `analogRead()`. It then checks whether the highest reading is below the night threshold. In night mode, both servos return to the central rest position and the program waits longer before checking again.

During normal operation, the code calculates two differential errors:

$$\Delta_H = (v_1 + v_3) - (v_2 + v_4), \qquad \Delta_V = (v_1 + v_2) - (v_3 + v_4).$$

If the absolute value of an error is greater than the tolerance, the corresponding servo moves by one degree. The horizontal servo is limited to 0°–180°, while the vertical servo is limited to 30°–150°. The Serial Monitor displays the four LDR readings and the current positions of both servos, which makes debugging easier.

The main source file can be stored in the repository as `solar_tracker.ino`.

3D modelling

The mechanical structure was designed as a 3D model before being printed. It includes:

- a fixed base;
- a support for the horizontal rotation;
- a moving arm for the second axis;
- a platform for the sensors and the future solar panel;
- mounting points for the two servo motors.

The model was adjusted to make the parts fit together and to allow the servos to rotate without blocking the structure. 3D modelling made it possible to check the dimensions and the position of the components before manufacturing the parts.

3D printing and assembly

After the modelling stage, the structure was produced using 3D printing. The printed parts were assembled with the servo motors and the electronic components. During assembly, we checked:

- that the servo motors were properly fixed;
- that both axes could move freely;
- that the parts were aligned;
- that the platform was stable;
- that the sensors and the future panel could be installed.

Some mechanical details were adjusted when necessary to improve the fit and movement of the printed parts.

Current status and limitations

The project successfully allowed us to work on Arduino programming, servo movement and the 3D design and fabrication of the tracker structure. The current prototype is focused on the mechanical movement and the control system.

The solar panel was not powered or tested for energy production. Therefore, the following aspects remain outside the scope of the current project: electrical output, efficiency, energy gain, and comparison with a fixed panel.

Possible next steps are:

- install the four LDRs on the printed sensor platform;
- test the response of the tracker to light from different directions;
- improve the panel mounting system;
- provide a stable power supply for the servos;
- synchronise the stored servo positions with the physical position after night mode;
- record sensor values and servo positions through the Serial Monitor.

Conclusion

This project combines Arduino programming, servo motor control, 3D modelling and 3D printing. The work completed so far concerns the movement of the two-axis mechanism and the construction of its support structure. A future version can add the LDR sensors and test the complete light-following function, but no energy-performance claims are made in this report.